

QUANTUM BASICS • BEGINNER NOTES

Sections 05–06

GUIDED LESSON • OBSERVE & CREATE

Measurement & H Gate

Learn what measurement actually tells us about a qubit, then use the Hadamard gate to create a state with equal chances of producing 0 or 1.

05 What a Measurement Tells You

Reading the result of a quantum state

1. What is measurement?

A **measurement** is the process of extracting a classical result from a quantum system. For a single qubit, we commonly measure in the computational basis and obtain either **0** or **1**.

Think of measurement as asking a question

Before measurement, a qubit can be in a quantum state that contains information about both $|0\rangle$ and $|1\rangle$. Measurement asks: **“What result do I get when I observe this qubit in the computational basis?”**

2. What can a measurement return?

When a single qubit is measured in the computational basis, the classical output is either **0** or **1**. These are the two possible measurement outcomes.

Outcome 0

The measurement result is the classical bit **0**.

Outcome 1

The measurement result is the classical bit **1**.

3. Measuring $|0\rangle$

Suppose the qubit is definitely in the state $|0\rangle$. There is no uncertainty between 0 and 1 in this state. Therefore, measuring it produces 0 with certainty.

CIRCUIT 1 - MEASURE $|0\rangle$



Measuring $|0\rangle$ gives the classical result 0.

$$P(0) = 100\% \quad P(1) = 0\%$$

4. Measuring $|1\rangle$

Similarly, if the qubit is definitely in $|1\rangle$, measurement produces the classical result 1 with certainty.

CIRCUIT 2 - MEASURE $|1\rangle$



Measuring $|1\rangle$ gives the classical result 1.

$$P(0) = 0\% \quad P(1) = 100\%$$

5. Measurement becomes interesting with superposition

The interesting case occurs when a qubit is in a **superposition**. A general single-qubit state can be written as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Here, α and β are called probability amplitudes. They describe the quantum state and determine the probabilities of the possible measurement outcomes.

6. From amplitudes to probabilities

To find the probability of each measurement result, we take the squared magnitude of the corresponding amplitude.

$$\text{Probability of 0: } P(0) = |\alpha|^2$$

$$\text{Probability of 1: } P(1) = |\beta|^2$$

Important: Amplitudes and probabilities are not the same thing. The amplitudes describe the quantum state, while their squared magnitudes give the probabilities of measurement outcomes.

7. Example: a 50–50 superposition

Consider the state: $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Both basis states have the same amplitude.

$$|+\rangle = 1/\sqrt{2} |0\rangle + 1/\sqrt{2} |1\rangle$$

$$P(0) = |1/\sqrt{2}|^2 = 1/2$$

$$P(1) = |1/\sqrt{2}|^2 = 1/2$$

Measurement probabilities

0		50%
1		50%

8. Does measurement always give the same answer?

No. If the qubit is in a superposition, repeated measurements can produce different outcomes. For a 50–50 state, one measurement might give 0, while another independent preparation and measurement might give 1.

EXAMPLE

Suppose we prepare the state $|+\rangle$ many times.

One possible sequence of measurement results could be: **0, 1, 1, 0, 1, 0, 0, 1**

The exact sequence is unpredictable, but over many repetitions the results tend toward approximately 50% zeros and 50% ones.

9. What happens to the state after measurement?

In the standard computational-basis measurement, once the measurement gives a definite result, the state is projected onto the corresponding basis state. So if the result is 0, the measured state becomes $|0\rangle$; if the result is 1, it becomes $|1\rangle$.

Result = 0

The state is projected to $|0\rangle$.

Result = 1

The state is projected to $|1\rangle$.

10. Measurement gives classical information

A quantum state contains quantum information. Measurement gives us a **classical result**. For one qubit measured in the computational basis, that classical result is a bit: either 0 or 1.

Quantum state → Measurement → Classical result

11. A complete measurement circuit

Let us put the idea together. We prepare a qubit in $|0\rangle$, apply no gate, and measure it.

CIRCUIT 3 · PREPARE → MEASURE



- 1 Start with a qubit in $|0\rangle$.
- 2 Do not apply any gate.
- 3 Measure the qubit.
- 4 The classical result is **0**.

KEY IDEA

Measurement tells you a classical outcome obtained from a quantum state. For a qubit measured in the computational basis, the possible results are **0 or 1**. The probabilities of those outcomes depend on the quantum state before measurement.

QUICK CHECK

1. What can a single computational-basis measurement give?
→ 0 or 1.
2. What happens when $|0\rangle$ is measured?

→ 0 with 100% probability.

3. What happens when $|1\rangle$ is measured?

→ 1 with 100% probability.

4. What determines the probability of measuring 0?

→ The squared magnitude of the $|0\rangle$ amplitude.

06 Create Equal Chances With H

Use the Hadamard gate to create a 50–50 state

1. Meet the H gate

The **Hadamard gate**, usually written as **H**, is one of the most important single-qubit gates. It is especially useful because it can transform a definite basis state into a superposition.

The big idea

If we start with $|0\rangle$ and apply H, we obtain a state in which measuring the qubit gives **0 or 1 with equal probability**. This creates a 50–50 quantum state.

2. The H gate matrix

Like other quantum gates, the Hadamard gate can be represented by a matrix.

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

3. H acting on $|0\rangle$

The most important beginner rule is:

$$H|0\rangle = |+\rangle$$

where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. This means the qubit is now in an equal superposition of $|0\rangle$ and $|1\rangle$.

4. The basic H circuit

Start with $|0\rangle$, apply H, and then measure.

CIRCUIT 4 · CREATE EQUAL CHANCES

$|0\rangle$



0 / 1

H transforms $|0\rangle$ into an equal superposition before measurement.

5. Why are the chances equal?

After applying H, the state is:

$$|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

The amplitude of $|0\rangle$ is $1/\sqrt{2}$, and the amplitude of $|1\rangle$ is also $1/\sqrt{2}$. Squaring each amplitude gives:

$$P(0) = |1/\sqrt{2}|^2 = 1/2$$

$$P(1) = |1/\sqrt{2}|^2 = 1/2$$

Result after H

0		50%
1		50%

6. Matrix calculation: $H|0\rangle$

We can verify the result using matrix multiplication. The state $|0\rangle$ is represented by:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$H|0\rangle$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= (|0\rangle + |1\rangle)/\sqrt{2}$$

7. What does H do to $|1\rangle$?

The Hadamard gate also transforms $|1\rangle$ into a superposition, but with a relative minus sign.

$$H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$$

Important: The minus sign does not mean that the probability of measuring 1 becomes negative. Probabilities are always non-negative. The sign is part of the quantum amplitude and can affect interference in later operations.

8. H creates superposition, not a random classical bit

It is tempting to say that H simply “randomly chooses 0 or 1”. That description is useful at a very basic level, but it misses an important point. H creates a **quantum superposition** first. The 50–50 probabilities appear when we measure that state.

Quantum state vs measurement result

After H: $(|0\rangle + |1\rangle)/\sqrt{2}$ is the quantum state. After measurement: **0 or 1** is the classical result.

9. Repeating the H experiment

A single run does not have to produce exactly half 0s and half 1s. Each individual measurement is probabilistic. If we repeat the experiment many times, however, the number of 0s and 1s should approach an approximately equal distribution.

EXAMPLE • 10 RUNS

One possible result: **0, 1, 0, 1, 1, 0, 1, 0, 1, 0**

Here we obtained: **5 zeros** and **5 ones**.

Another set of 10 runs could give 6 zeros and 4 ones. That is also perfectly possible. The 50–50 value describes the probability over many repetitions, not a guarantee about every small sample.

10. H followed by measurement

This is one of the most useful circuits to remember. It converts a definite $|0\rangle$ state into an equal superposition and then measures it.

CIRCUIT 5 • H THEN MEASUREMENT



- 1 Begin with the qubit in $|0\rangle$.
- 2 Apply the Hadamard gate H.
- 3 The state becomes $(|0\rangle + |1\rangle)/\sqrt{2}$.

- 4 Measure the qubit.
- 5 The result is 0 or 1, each with probability **50%**.

11. H is different from X

You have already learned the X gate. Comparing X and H makes their roles easier to remember.

Gate	Input	Output	Main idea
X	$ 0\rangle$	$ 1\rangle$	Flips the basis state
X	$ 1\rangle$	$ 0\rangle$	Flips the basis state
H	$ 0\rangle$	$(0\rangle + 1\rangle)/\sqrt{2}$	Creates equal superposition
H	$ 1\rangle$	$(0\rangle - 1\rangle)/\sqrt{2}$	Creates another superposition

12. H can also undo itself

An interesting property of the Hadamard gate is that applying H twice returns the qubit to its original state.

CIRCUIT 6 - H FOLLOWED BY H



The first H creates superposition; the second H returns the state to $|0\rangle$.

$$H^2 = I$$

So if we start with $|0\rangle$:

$$|0\rangle \rightarrow H \rightarrow |+\rangle \rightarrow H \rightarrow |0\rangle$$

13. Why is H important in quantum computing?

The Hadamard gate is extremely useful because it allows us to move between computational-basis states and superposition states. Many quantum algorithms begin by applying H gates to qubits to create a large space of possible states.

Before H

The qubit is in a definite computational-basis state, such as $|0\rangle$.

After H

The qubit can be in a superposition involving both $|0\rangle$ and $|1\rangle$.

14. The complete picture

We can now connect the previous topics. A no-gate circuit leaves the state unchanged. X flips the computational basis state. H creates a superposition. Measurement converts the quantum state into a classical result.

Operation	What it does	Example
No gate	Leaves the state unchanged	$ 0\rangle \rightarrow 0\rangle$
X	Flips $ 0\rangle$ and $ 1\rangle$	$ 0\rangle \rightarrow 1\rangle$
H	Creates superposition	$ 0\rangle \rightarrow +\rangle$
Measurement	Produces a classical result	$ +\rangle \rightarrow 0 \text{ or } 1$

KEY IDEA

The Hadamard gate H takes a definite basis state and creates a superposition. In particular, $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. When this state is measured in the computational basis,

the result is **0 with 50% probability** and **1 with 50% probability**. Measurement then converts that quantum information into a classical result.

QUICK CHECK

1. What does H do to $|0\rangle$?

→ $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

2. What are the measurement probabilities after $H|0\rangle$?

→ 50% for 0 and 50% for 1.

3. Does H directly produce a classical random bit?

→ No. It first creates a quantum superposition; measurement produces the classical result.

4. What does H do to $|1\rangle$?

→ $H|1\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$.

5. What happens if H is applied twice?

→ $H^2 = I$, so the original state is restored.

6. What does measurement tell us?

→ It gives a classical outcome such as 0 or 1, according to the probabilities determined by the quantum state.